

CASSA RA (Astrophysics) 2026-27 — Practice Sheet

Exam: 16 August 2026, 11:00 am · CASSA, IUB (BC Rooftop, Level-8) Written 11:00-11:45 · Viva from 12:00

How to use: Written test = "scientific background related to the position" (supernova cosmology, SED fitting, general astro). Viva = "about what you submitted" (your 3-paper SN project, RGC, PIE-ADA, knackbook, background). Read Section 1-3 for concepts, Section 4 to rehearse your own work out loud, Section 5 numbers by heart, Section 6 self-test.

SECTION 1 — Cosmology fundamentals (written)

Hubble's law. Recession velocity $v = H_0 d$. $H_0 \approx 70$ km/s/Mpc (tension: ~ 67 CMB/Planck vs ~ 73 local/SN). Redshift $z = (\lambda_{\text{obs}} - \lambda_{\text{emit}})/\lambda_{\text{emit}}$; for small z , $v \approx cz$.

Expanding universe / FLRW. Space described by scale factor $a(t)$, $a(\text{now})=1$, $1+z = 1/a$. Friedmann equation links expansion rate to energy density: $H^2 = (8\pi G/3)\rho - k/a^2 + \Lambda/3$. Components: matter ($\Omega_m \approx 0.3$), dark energy ($\Omega_\Lambda \approx 0.7$), radiation (tiny), curvature (≈ 0 , flat).

Dark energy. Drives accelerated expansion. Equation of state $w = P/\rho$; cosmological constant Λ has $w = -1$. Acceleration needs $w < -1/3$.

Discovery of acceleration (1998-99). Riess et al. 1998 (High- z team) + Perlmutter et al. 1999 (SCP) used distant Type Ia SNe $\rightarrow$ they were **fainter than expected** in a decelerating universe $\rightarrow$ universe is accelerating $\rightarrow$ Nobel 2011. This is exactly the result your project's "opponent" (Son et al.) tries to weaken.

Distances. - Distance modulus: $\mu = m - M = 5 \log_{10}(d_L / 10 \text{ pc})$. - Luminosity distance d_L , comoving distance, angular-diameter distance $d_A = d_L/(1+z)^2$. - SN Ia give $d_L \rightarrow$ build the **Hubble diagram** (μ vs z) $\rightarrow$ fit cosmology.

Standard candle vs standardizable. A true standard candle has fixed M . SN Ia are **standardizable**: raw scatter reduced using light-curve shape and color.

SECTION 2 — Type Ia supernovae & standardization (written + viva core)

Progenitor. Carbon-oxygen **white dwarf** in a binary, accreting to near the **Chandrasekhar mass** $\approx 1.4 M_\odot$, then thermonuclear runaway. Two channels: **single-degenerate** (WD + normal companion) and **double-degenerate** (WD + WD merger). Uniform trigger mass $\Rightarrow$ uniform peak luminosity $\Rightarrow$ good candles.

Why standardizable — Phillips relation (1993). Brighter SN Ia decline **more slowly**; light-curve width $\propto$ luminosity. Correct using: - **stretch** / Δm_{15} (light-curve width), coefficient α - **color** (reddening/intrinsic), coefficient β - Tripp formula: $\mu = m_B - M + \alpha \cdot x_1 - \beta \cdot c$.

Light-curve fitters. SALT2 (x_1 , c), MLCS2k2, SNooPy (used for CSP near-IR).

Host-galaxy correlations (the heart of your project). Even after stretch+color, residuals correlate with the host: - **Mass step**: SN in high-stellar-mass hosts appear ~ 0.05 - 0.06 mag brighter after correction. A step is added at $\sim 10^{10} M_\odot$. - **Age step / environment**: some argue the *real* driver is host stellar **age** (older population $\rightarrow$ brighter residual). - **Why it matters**: if host properties evolve with redshift (high- z hosts are younger), an uncorrected age bias could **mimic** the dark-energy signal. That is Son et al.'s (2025) claim — that this bias is big enough to question acceleration.

Your refutation logic (say this clearly in viva): 1. Son et al. use optical-only ages $\rightarrow$ weak age constraints. 2. You use **CSP-I 7-band ugriYJH** (optical + near-IR) $\rightarrow$ near-IR breaks the age-dust-metallicity degeneracy $\rightarrow$ tighter ages. 3. You use **global** host age (whole galaxy), not just local. 4.

Result: age step $+0.032 \pm 0.064$ mag = **0.50σ** , consistent with zero → no significant age bias → acceleration stands.

SECTION 3 — SED fitting, Bayesian inference & samplers (written + viva core)

SED = Spectral Energy Distribution — galaxy flux across wavelength bands. Fit a model to observed photometry to recover **stellar mass, age, star-formation history (SFH), metallicity, dust**.

Stellar population synthesis (SPS). Combine simple stellar populations (single age+metallicity) weighted by an SFH and an **IMF** (initial mass function, e.g. Chabrier/Kroupa) → predicted SED. Codes: **FSPS** (used in Rose et al. 2019 pipeline you built on), BC03, Prospector, BAGPIPES.

Bayesian inference. Posterior $\propto$ Likelihood $\times$ Prior $\rightarrow P(\theta|D) \propto P(D|\theta) P(\theta)$. - θ = parameters (mass, age, dust...). Likelihood usually Gaussian: $\ln L = -\frac{1}{2} \sum (\text{data} - \text{model})^2 / \sigma^2$. - Report **posterior median + credible interval**, not just best-fit. - **Evidence** $Z = \int L(\theta) \pi(\theta) d\theta$ — normalizing integral; nested samplers give it for free (good for model comparison).

Samplers you compared (know the difference cold): | Sampler | Type | One-liner | |---|---|---| | **emcee** | MCMC (affine-invariant ensemble) | Many "walkers" explore together; robust but slow to converge; no evidence. | | **MultiNest** | Nested sampling | Ellipsoidal decomposition; handles **multimodal** posteriors well; gives evidence. | | **dynesty** | Dynamic nested sampling | Pure-Python nested sampling; allocates live points where needed. | | **Nautilus** | Neural-network-boosted importance nested sampling | Learns the posterior shape → far fewer likelihood calls → **~60x faster than emcee** here. |

Your Paper 1 result: identical likelihood/priors/FSPS model, only the sampler changes. Nautilus $\approx$ **60x faster** than emcee, ages offset by **$+0.64 \pm 0.29$ Gyr (2.21σ)** → sampler choice is itself a systematic. Paper 2: apply Nautilus (MultiNest fallback for bimodal posteriors) to CSP-I. Paper 3: `snhostage` pipeline + `CANDLE` Streamlit tool.

MCMC health checks (likely viva): convergence (Gelman-Rubin $\hat{R} \rightarrow 1$, autocorrelation time), burn-in, chain mixing, effective sample size.

SECTION 4 — VIVA: rehearse your own work (say each aloud)

A. "Explain your supernova project in one minute."

I test whether Type Ia supernova host-galaxy stellar age biases their standardized brightness — a claim by Son et al. 2025 that, if true, would undermine the evidence for cosmic acceleration. I do Bayesian SED fitting of host galaxies to get their ages, then look for a step in Hubble residuals with age. Three papers: (1) a sampler-systematics study showing sampler choice alone shifts ages; (2) an independent test on CSP-I with 7-band photometry finding the age step is consistent with zero; (3) the open pipeline `snhostage` plus a web tool.

B. Likely drill-downs — have a crisp answer: - *Why is Nautilus faster than emcee?* It uses neural-network importance nested sampling to learn the posterior, so it needs far fewer likelihood evaluations; emcee must random-walk to convergence. - *If samplers disagree, which is right?* They share likelihood/priors/model, so differences are **method systematics**. Cross-check with evidence, posterior predictive, and independent samplers; that's the point of Paper 1 — quantify, don't hide it. - *Why does 7-band `ugriYJH` beat optical-only?* Near-IR is dominated by old stars and less sensitive to dust → breaks the age-dust-metallicity degeneracy → tighter age. - *What does "age step = $+0.032 \pm 0.064$ mag, 0.50σ " mean?* The brightness difference between old- and young-host SN is 0.032 mag, but its error (0.064) is twice as big → statistically indistinguishable from zero → no evidence for the bias. - *Global vs local age — why global?* You measure the whole host's stellar population, which is more robust for these samples than a small local aperture; ties to the physical driver Son et al. invoke. - *What is the mass step and how is it related?* Residual correlation with host stellar mass (~ 0.05 mag at $10^{10} M_{\odot}$); age is a

proposed underlying cause. Show you know both. - *Priors* — *what did you assume?* (Be ready: SFH form, age/metallicity/dust ranges, redshift fixed to spectroscopic.) State them plainly and say why.

C. RGC (radio AGN, arXiv:2510.22190, accepted A&A 2026).

*I did data curation and catalogue preparation — cross-matching and source validation — for the training set of a semi-supervised CNN that classifies radio-AGN morphology. For v2 I guided an image-annotation tool through testing. - Know: **Fanaroff-Riley I vs II** (FRI = edge-darkened, core-bright, lower power; FR II = edge-brightened hotspots, higher power). Radio emission = **synchrotron** from relativistic electrons in magnetic fields. Ties into the I-HOW Radio Astronomy workshop you were selected for.*

D. PIE-ADA (undergrad thesis, IEEE QPAIN 2026).

Physics-Informed Ensemble with Adaptive Data Augmentation for 14-class photometric transient classification for LSST. On PLAsTiCC (7,848 objects) I added physically motivated augmentations — cosmological time dilation, wavelength-dependent dust extinction, correlated noise — and 271 multi-scale features across 6 passbands. LightGBM: weighted log-loss 0.5763 ± 0.0083 , 80.33% accuracy, 24-49% better than baselines. - Why physics-informed augmentation? Real surveys see redshifted, dust-reddened, noisy light curves; augmenting with the actual physics teaches invariances instead of random noise. - Connects directly to SN work: same light-curve + transient-classification toolkit.

E. Background / motivation (viva always asks). - *Why astrophysics / why CASSA?* Olympiad path (12th BdAO, IOAA 2017 Thailand), self-taught thesis, joined CASSA for the research you care about; want to finish the SN host-age analysis and build toward a PhD in supernova cosmology / computational astrophysics with Dr. Uddin (time-domain, host demographics, statistics→cosmology). - *Recent selection:* IAU Hands-on Workshop in Radio Astronomy + 10th Byurakan Intl Summer School, Armenia, Sep 2026 — your first international astronomy selection since IOAA 2017. - *knackbook:* a social platform (any creative/intellectual field) with real-time chat; shows your software side (JavaScript, Firebase, GitHub Pages). - *Weakness / what's incomplete?* Honest: the host-age analysis isn't finished — that's exactly what the RA year is for.

SECTION 5 — Numbers & names to memorize (flash-card)

- $H_0 \approx 70 \text{ km/s/Mpc}$ · $\Omega_m \approx 0.3$, $\Omega_\Lambda \approx 0.7$, flat · dark energy $w = -1$ for Λ .
- Chandrasekhar mass $\approx 1.4 M_\odot$ · SN Ia = C-O white dwarf, thermonuclear.
- Phillips: brighter $\Rightarrow$ slower decline · Tripp: $\mu = m_B - M + \alpha x_1 - \beta c$.
- Mass step $\approx 0.05\text{-}0.06 \text{ mag}$ at $\sim 10^{10} M_\odot$.
- Acceleration: Riess 1998 + Perlmutter 1999, Nobel 2011.
- Bayes: Posterior $\propto$ Likelihood $\times$ Prior; evidence $Z = \int L \pi d\theta$.
- Your Paper 1: Nautilus $\sim 60\times$ faster than emcee; age offset **$+0.64 \pm 0.29 \text{ Gyr}$ (2.21σ)**.
- Your Paper 2: CSP-I age step **$+0.032 \pm 0.064 \text{ mag}$ (0.50σ)**, consistent with zero $\rightarrow$ refutes Son et al. 2025.
- 5 sampler configs: 2 $\times$ emcee, MultiNest, dynesty, Nautilus · pipeline: FSPS (Rose et al. 2019), SDSS-II hosts.
- PIE-ADA: 14 classes, PLAsTiCC 7,848 objects, 271 features, 6 bands, LightGBM log-loss **0.5763 ± 0.0083 , 80.33% acc.**
- FRI (edge-dark) vs FR II (edge-bright hotspots); radio = synchrotron.

SECTION 6 — Self-test (cover the answers above)

Written-style: 1. Write Hubble's law and give H_0 . What is the H_0 tension? 2. Why are SN Ia standardizable, not standard, candles? Name the two correction terms. 3. What is the Chandrasekhar mass and why does it make SN Ia uniform? 4. State Bayes' theorem for parameter estimation. What is the evidence and which samplers give it? 5. What did Riess 1998 / Perlmutter 1999 discover, and from what? 6. Define luminosity distance and distance modulus. 7. FRI vs FR II — describe the morphology difference. What radiation mechanism?

Viva-style (answer out loud, 60-90 s each): 8. Explain your whole SN project and what claim you are testing. 9. Why does host-galaxy age matter for measuring dark energy? 10. Why is your CSP-I 7-band result stronger than Son et al.'s optical-only? 11. Interpret " $+0.032 \pm 0.064$ mag, 0.50σ ." 12. Why is Nautilus $\sim 60\times$ faster than emcee? 13. If two samplers give different ages, how do you decide? 14. Summarize PIE-ADA and how it connects to your SN work. 15. Why CASSA, why astrophysics, and what do you want from the RA year?

Exam-day tips

- Bring your printed CV/SOP; skim them in the morning — viva is "about what you submitted", so **every line on your CV is fair game**.
- If you don't know something: say what you *do* know, reason from basics, don't bluff.
- Written: define terms, write the key equation, one clean sentence of meaning.
- Speak the numbers in Section 5 confidently — that specificity is what impresses a viva panel.

Good luck, sir. Apni prepared achen — just rehearse Section 4 out loud once or twice.